# Writing results and limitations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

## Teach the evidence-to-conclusion chain for Writing results and limitations

Keep counts, outcome definitions and failed conditions visible without substituting unmeasured benefits.

**Keep one habit:** Model the main case → reveal the picture gradually → probe limits → assess independent reasoning → plan a changed-case return.

MAIN LESSON · SECTION 1

## Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

### Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of writing results and limitations

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

## Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

## Opening retrieval

Without opening the old lesson, explain one key idea from Writing the methods section. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of writing the methods section

Reference example, not the only acceptable wording: Teacher explanation: A methods section should let a reader reconstruct the design and computation. Detail should explain choices that affect **evidence**.

Report question, units, sampling, permissions, data roles, baseline, procedure, metrics, uncertainty assumptions, missing/exclusion handling and versions. Distinguish prespecified and later changes.

Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of negative results and claim revision

Reference example, not the only acceptable wording: Teacher explanation: Negative results report that the **evidence** did not meet a declared expectation.

They can reveal design limits or useful constraints without proving no effect anywhere. Compare results with the original criterion and uncertainty assumptions.

Preserve the failed hypothesis, discuss plausible **mechanism**s and label revised questions as new work.

Without opening the old lesson, explain one key idea from Running the primary experiment. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of running the primary experiment

Reference example, not the only acceptable wording: Teacher explanation: The primary experiment follows the frozen plan and captures every eligible outcome. Unexpected failures are recorded rather than silently retried away.

Run declared cases, store raw paired results and deviations, and keep primary analysis separate from exploratory repairs. Stop or escalate only according to the stated rule.

## Pictured end goal

Explain writing results and limitations using the records and boundaries of this fictional investigation.

### ▼ Earlier wording, alternatives and picture transcripts

#### Teach the learning cycle

#### Retrieval answer guidance

#### Writing the methods section

#### Negative results and claim revision

#### Running the primary experiment

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. Results and limitations should state observed estimates before interpreting them. Separate measured comparisons from causal or deployment claims.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

#### Outcomes

Explain the **mechanism** and terminology of writing results and limitations

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Preparation

### Prerequisite answer

### End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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MAIN LESSON · SECTION 2

# Model the whole case and check each inference

> Keep the independent answer private until after the first attempt.

## Definitions and complete contract

Results and limitations should state observed estimates before interpreting them. Separate measured comparisons from causal or deployment claims.

Give counts, denominators, effect sizes, valid uncertainty and failed conditions. State missing **evidence** and preserve the original primary outcome.

| Results item | Complete supplied exercise contract |
|---|---|
| Main supplied counts | A eight of twelve passes; B nine of twelve |
| Main failures | All three B failures concern conflicting sources |
| Missing conflict denominator | Total number of conflict cases not supplied; no conflict-slice pass rate can be calculated |
| Main user evidence | No user study occurred |
| Separate picture | A supports nine of twelve answers; B ten of twelve, one approved collection |
| Unmeasured picture outcome | User learning is not recorded |

| Results item | Complete supplied exercise contract |
|---|---|
| Unknown inference | Significance, stable population effect, cause, timings and deployment usefulness not supplied |

## Main evidence ledger and checked key

| Main results component | Supported statement |
|---|---|
| Observed pass rates | A 8/12 ≈ 66.667%; B 9/12 = 75% |
| Descriptive difference | 1/12 ≈ 8.333 percentage points in B's favor |
| Failure category | Three B failures involve conflicting sources |
| What category statement does not imply | Not that every conflict case failed; total conflict cases unknown |
| User-study status | No study occurred; classroom usefulness not measured |
| Paired identities / statistical effect | Not supplied; no reconstructed transitions or significance |
| Scoped report | B passed one more of twelve supplied cases, with conflict failures retained |

Main A 8/12 ≈ 66.667%, B 9/12 = 75%, gain 1/12 ≈ 8.333 percentage points. All three B failures concern conflicting sources, but total conflict cases are unknown, so no conflict-slice rate follows. No user study supports classroom usefulness. Separate picture A support 9/12 = 75%, B 10/12 ≈ 83.333%, same one-case gain but a different stated outcome. No learning or timing claim follows.

Independent candidate 10/15 ≈ 66.667%, baseline 9/15 = 60%, descriptive gain 1/15 ≈ 6.667 points. State no timings measured and no established population/significance or broad task generalization. Accept other concrete supported limits without inventing missing observations.

### Writing results and limitations — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| B passes 9/12, A 8/12; three B failures all concern conflicting sources. No user study occurred. | Give counts, denominators, effect sizes, valid uncertainty and failed conditions. State missing evidence and preserve the original primary outcome. | 75% versus 66.667%, gain 8.333 points on 12 cases; conflict failures remain. No classroom usefulness or universal reliability claim is supported. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: 75% versus 66.667%, gain 8.333 points on 12 cases; conflict failures remain. No classroom usefulness or universal reliability claim is supported. Qualification: A limitation should directly bound the corresponding claim rather than contradict an oversized headline. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

B passes 9/12, A 8/12; three B failures all concern conflicting sources. No user study occurred.

75% versus 66.667%, gain 8.333 points on 12 cases; conflict failures remain. No classroom usefulness or universal reliability claim is supported.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

## Boundary and misconception probes

| Results/limits audit | Checked interpretation |
|---|---|
| One more observed pass | Report magnitude and denominator |
| All B failures involve conflict | Failure category known; total conflict slice unknown |
| No user study | Do not claim measured classroom usefulness |
| More supported answers | Not measured learning or general intelligence |
| Higher descriptive count | No automatic significance or causal inference |
| No timings measured | Do not claim speed improvement |
| Aggregate paired counts only | Do not invent exact case transitions |
| A limitation is acknowledged | Does not license keeping the unsupported broad claim |

## Separate pictured facts and full reasoning

Write results without enlarging them. A fictional study evaluates answer support on twelve fixed questions from one approved collection.

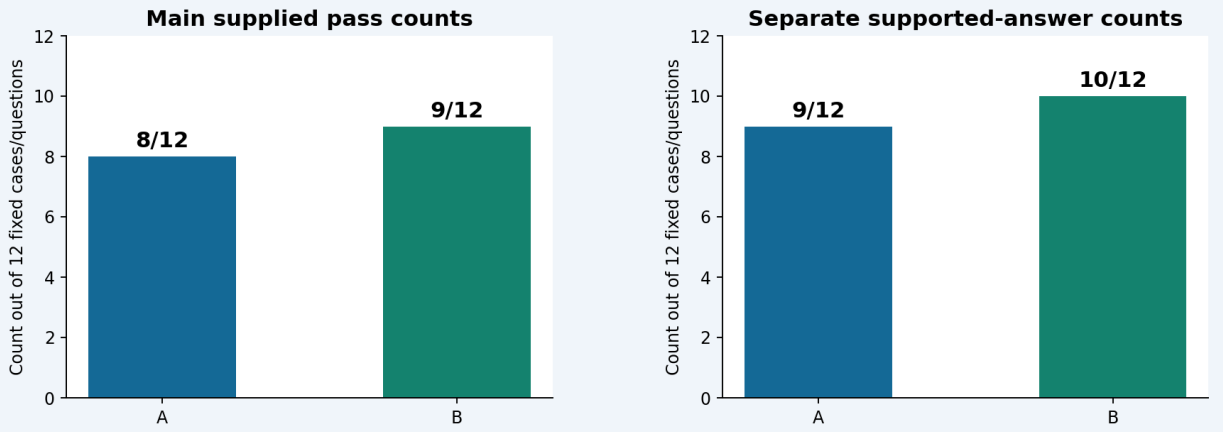
A supports 9/12 answers; B supports 10/12. No user learning outcomes are recorded.

The separate picture measures answer support on twelve fixed questions from one approved collection. A supports nine answers, or 75%; B supports ten, or approximately 83.333%. The difference is one supported answer, about 8.333 percentage points. This is a support outcome, not evidence that any learner became smarter. No user learning, timings, significance or stable population effect is supplied.

Keep the main eight-versus-nine pass case separate from this nine-versus-ten support case. A broader question about learners would require its own appropriate design and evidence, not a stronger adjective in the conclusion.

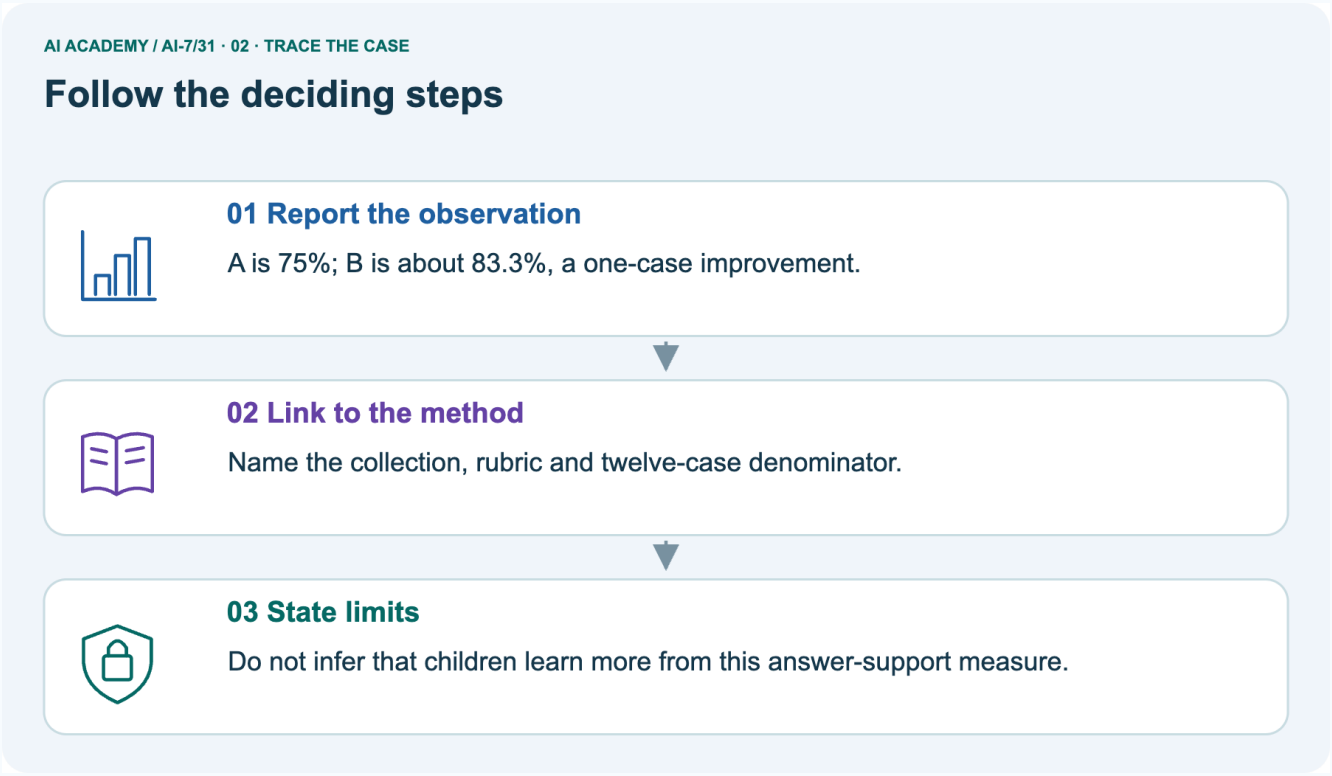
| Separate support outcome | A | B |
|---|---|---|
| Supported answers | 9/12 | 10/12 |
| Observed rate | 75% | ≈83.333% |
| Difference | One additional supported answer | ≈8.333 percentage points |
| Collection | Same stated approved collection | Only this fixed question set |
| Learner improvement | Not measured | Not measured |
| Timing / significance | Not supplied | Not supplied |
| Claim every student becomes smarter | Unsupported outcome substitution | Not a consequence of support counts |

## Describe the measured outcome before making a broader claim




Each comparison differs by one of twelve cases, about 8.333 percentage points.
No user learning or timings were measured. The two scenarios have different supplied outcomes.

## Picture trace and changed-condition key




Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Report the observation | A is 75%; B is about 83.3%, a one-case improvement. |
| 2. Link to the method | Name the collection, rubric and twelve-case denominator. |
| 3. State limits | Do not infer that children learn more from this answer-support measure. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Results describe what was measured; limitations prevent substitution of a different outcome or population. A small descriptive improvement needs its scope attached.

No significance or stable population effect is established; broader task and learner **evidence** would require another study.

Change: Write "B makes every student smarter".

**Prediction**: Remove that sentence as unsupported by the study.

Reason: The measured unit was answers, not student understanding or intelligence.

### ▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

| Term | Meaning |
| --- | --- |
| Descriptive estimate | A quantity calculated from observed data |
| Interpretation | A statement about what results establish |
| Limitation linkage | Connecting each restriction to the claim it limits |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

| Thinking move | Teacher prompt |
| --- | --- |
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

## A pictorial worked case: Writing results and limitations

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/31

# Writing results and limitations

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Report the observation

A is 75%; B is about 83.3%, a one-case improvement.



## 2. Link to the method

Name the collection, rubric and twelve-case denominator.



## 3. State limits

Do not infer that children learn more from this answer-support measure.

**CHANGE ONE THING**

Write "B makes every student smarter".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

**Why this result follows**

**What this example does not establish**

**Change one thing in the pictured case**

**Ask while showing the picture**

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

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MAIN LESSON · SECTION 3

# Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

## Current completion task and checked reasoning

Completion task: B passes 9/12, A 8/12; three B failures all concern conflicting sources. No user study occurred. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: 75% versus 66.667%, gain 8.333 points on 12 cases; conflict failures remain. No classroom usefulness or universal reliability claim is supported.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual misconception check

Use the worked case for Writing results and limitations. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A limitation should directly bound the corresponding claim rather than contradict an oversized headline.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Pictured reasoning and changed case

Compute the supported-answer rates for 9/12 and 10/12. What actual outcome was measured, and why is a claim that every student becomes smarter outside this **evidence**?

A is 75%; B is about 83.3%, a one-case improvement. Name the collection, rubric and twelve-case denominator. Do not infer that children learn more from this answer-support measure.

Changed case: Remove that sentence as unsupported by the study. The measured unit was answers, not student understanding or intelligence.

Changed-case reasoning: Remove that sentence as unsupported by the study. The measured unit was answers, not student understanding or intelligence.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

## Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

## ▼ Earlier wording, alternatives and picture transcripts

| Time or stage | Teacher action |
|---|---|
| Session 1: 0–7 minutes | Retrieve the prerequisite; identify the input, procedure and evidence source. |
| 7–18 minutes | Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation. |
| 18–32 minutes | Trace the declared procedure and worked case. Complete W2, recording intermediate steps. |
| 32–45 minutes | Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations. |
| Session 2: 0–8 minutes | Retrieve the worked reasoning and collect the guided response before discussing it. |
| 8–20 minutes | Compare a proposed boundary variant with the original case; identify what remains fixed. |
| 20–35 minutes | Collect W4 independently. Require the procedure trace and the bounded conclusion. |
| 35–45 minutes | Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts. |

### Reduce help deliberately

### Individual reasoning check

### Pictorial practice question

### Reasoned teacher answer

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MAIN LESSON · SECTION 4

# Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

## Complete inputs,expected results and limits

Main A 8/12 ≈ 66.667%, B 9/12 = 75%, gain 1/12 ≈ 8.333 percentage points. All three B failures concern conflicting sources, but total conflict cases are unknown, so no conflict-slice rate follows. No user study supports classroom usefulness. Separate picture A support 9/12 = 75%, B 10/12 ≈ 83.333%, same one-case gain but a different stated outcome. No learning or timing claim follows.

Independent candidate 10/15 ≈ 66.667%, baseline 9/15 = 60%, descriptive gain 1/15 ≈ 6.667 points. State no timings measured and no established population/significance or broad task generalization. Accept other concrete supported limits without inventing missing observations.

## Diagnose the current misconceptions

The worked result establishes every future case: A limitation should directly bound the corresponding claim rather than contradict an oversized headline.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

## Choose accessible support

| What the response shows | Next teaching action |
| --- | --- |
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |

| What the response shows | Next teaching action |
| --- | --- |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

**Match the support to the first error**

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MAIN LESSON · SECTION 5

## Assess the new research scenario

> Keep the selected answer private until after a first attempt.

### Selected independent task and checked answer

Selected case: Candidate 10/15, baseline 9/15; no timings measured. Write result and two limits.

Teacher explanation: 66.667% versus**60%**, gain6.667 points in15 cases. Timing is unmeasured and coverage/population generalisation remain limited.

## Judge explanation,trace and limits

| Dimension | Evidence required |
| --- | --- |
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

## Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

## W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Results and limitations should state observed estimates before interpreting them. Separate measured comparisons from causal or deployment claims.

Give counts, denominators, effect sizes, valid uncertainty and failed conditions. State missing **evidence** and preserve the original primary outcome.

## W2 Trace the worked case

B passes 9/12, A 8/12; three B failures all concern conflicting sources. No user study occurred. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: 75% versus 66.667%, gain 8.333 points on 12 cases; conflict failures remain. No classroom usefulness or universal reliability claim is supported.

## W3 Audit the limit

A report omits this qualification: A limitation should directly bound the corresponding claim rather than contradict an oversized headline.

Explain how omitting it could change a reader's **interpretation**, and write an accurate replacement.

Teacher explanation: A limitation should directly bound the corresponding claim rather than contradict an oversized headline.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

## W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A limitation should directly bound the corresponding claim rather than contradict an oversized headline.

The new case should test this particular **mechanism** rather than merely restate the original result.

### ▼ Earlier wording, alternatives and picture transcripts

#### Guided and exit questions

Guided: 75% versus 66.667%, gain 8.333 points on 12 cases; conflict failures remain. No classroom usefulness or universal reliability claim is supported. Qualification: A limitation should directly bound the corresponding claim rather than contradict an oversized headline.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

### W1 Explain the mechanism

### W2 Trace the worked case

### W3 Audit the limit

### W4 Apply a new case independently

Candidate 10/15, baseline 9/15; no timings measured. Write result and two limits.

Teacher explanation: 66.667% versus**60%**, gain6.667 points in15 cases. Timing is unmeasured and coverage/population generalisation remain limited.

### W5 Design a revealing follow-up

### Independent assessment and fresh retry

W4: 66.667% versus**60%**, gain6.667 points in15 cases. Timing is unmeasured and coverage/population generalisation remain limited. Require intermediate reasoning and the stated scope. Fresh retry: Candidate7/10, baseline8/10.

Expected reasoning: **70%** versus**80%**, candidate lower10 points; report negative result and sample scope. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

### Assess independent transfer

### Selected case answer

### Feedback that the learner uses

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MAIN LESSON · SECTION 6

## Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

## Project and next-week bridge

Add a labelled writing results and limitations **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of writing results and limitations.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Peer review and response. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

## Delayed return and support record

Suggested return points in this level: ai-7/32, ai-7/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
| --- | --- |
| Learner code and date | |
| Concept and case used; prior exposure | |

| Record | Teacher evidence |
| --- | --- |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### ▼ Earlier wording, alternatives and picture transcripts

### Plan the delayed return

### Project evidence from this case

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

## Reference · Alternative planning checklist

> Use alongside the main route when helpful.

### Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of writing results and limitations

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: B passes 9/12, A 8/12; three B failures all concern conflicting sources. No user study occurred. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: 75% versus 66.667%, gain 8.333 points on 12 cases; conflict failures remain. No classroom usefulness or universal reliability claim is supported.

Check every learner before discussion: Use the worked case for Writing results and limitations. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A limitation should directly bound the corresponding claim rather than contradict an oversized headline.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Write "B makes every student smarter". Require a **prediction** and a reason before comparing. Expected result: Remove that sentence as unsupported by the study.

Reason: The measured unit was answers, not student understanding or intelligence.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

# Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

## Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

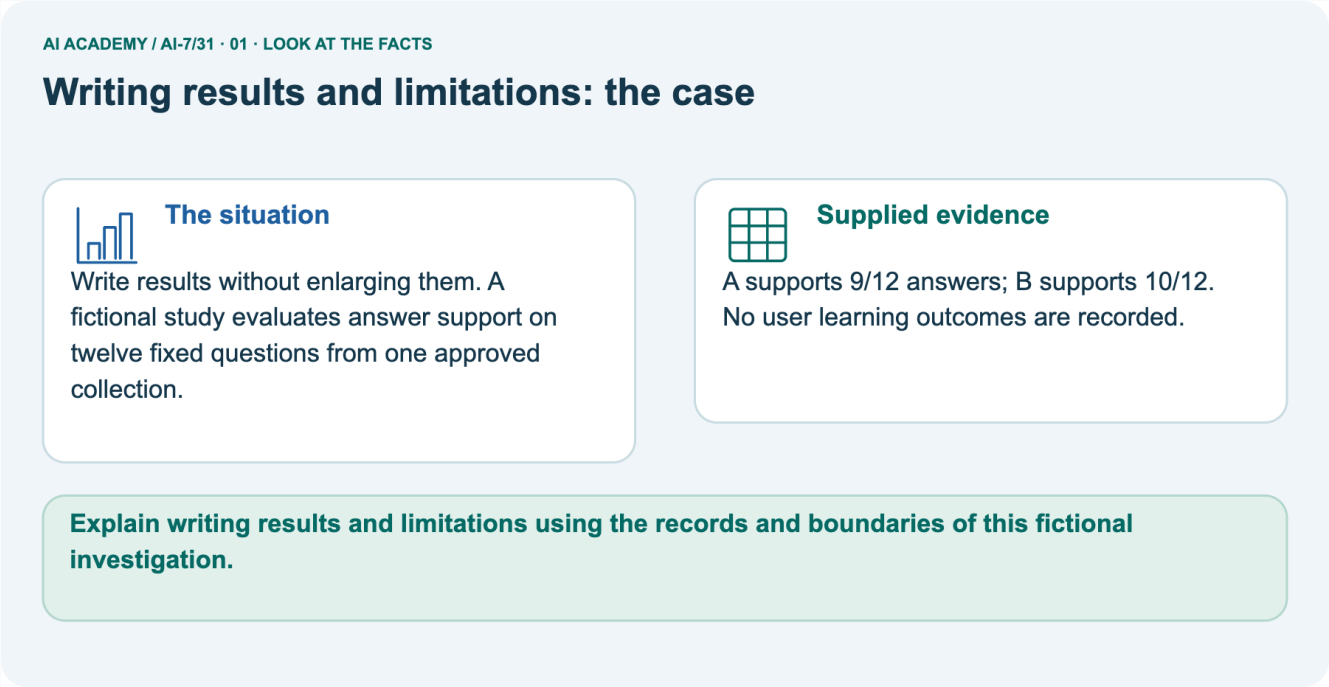
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

## Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

## 01. Writing results and limitations: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

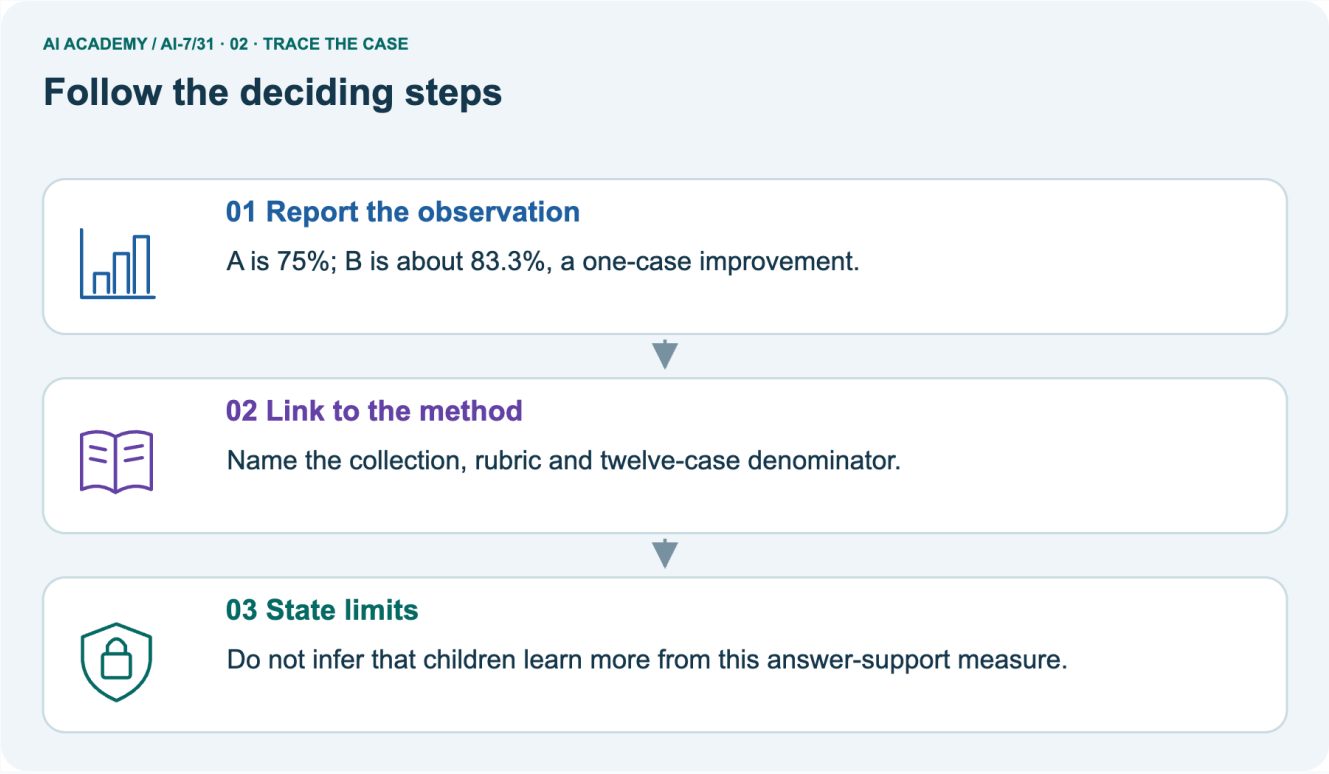
Write results without enlarging them. A fictional study evaluates answer support on twelve fixed questions from one approved collection. Supplied facts: A supports 9/12 answers; B supports 10/12.

No user learning outcomes are recorded.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 02. Follow the deciding steps



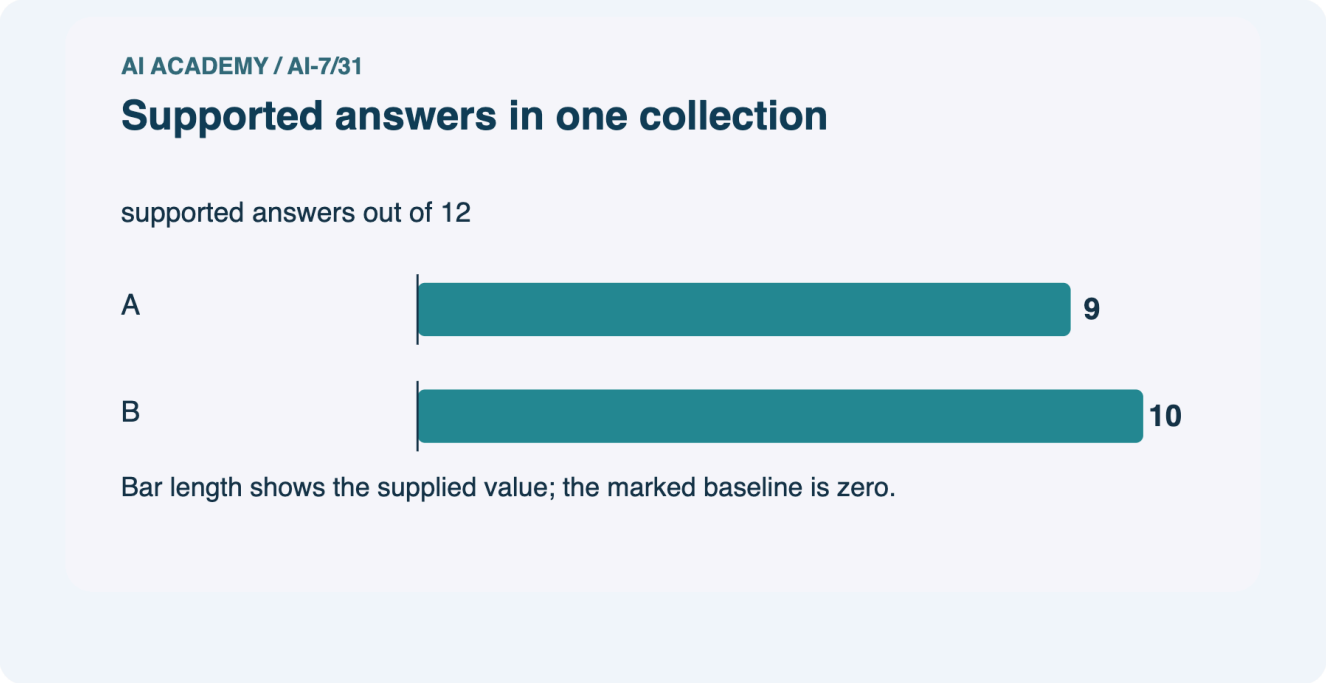

Follow the labelled picture with the matching explanation. Open the vector diagram

Report the **observation**: A is 75%; B is about 83.3%, a one-case improvement. → Link to the method: Name the collection, rubric and twelve-case denominator. → State limits: Do not infer that children learn more from this answer-support measure.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

## 03. Supported answers in one collection




Follow the labelled picture with the matching explanation. Open the vector diagram

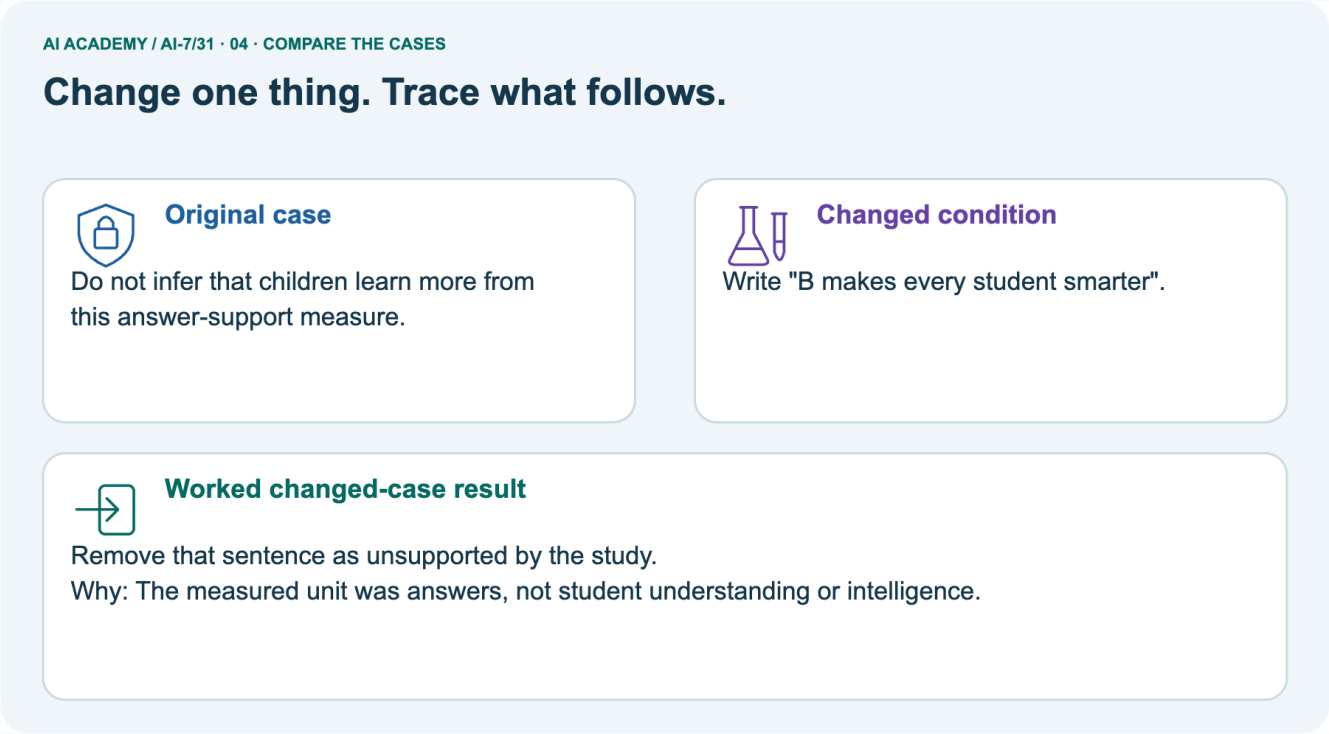
Supported answers in one collection. Units: supported answers out of 12. A: 9. B: 10.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Do not infer that children learn more from this answer-support measure. Changed condition: Write "B makes every student smarter". Expected result: Remove that sentence as unsupported by the study.

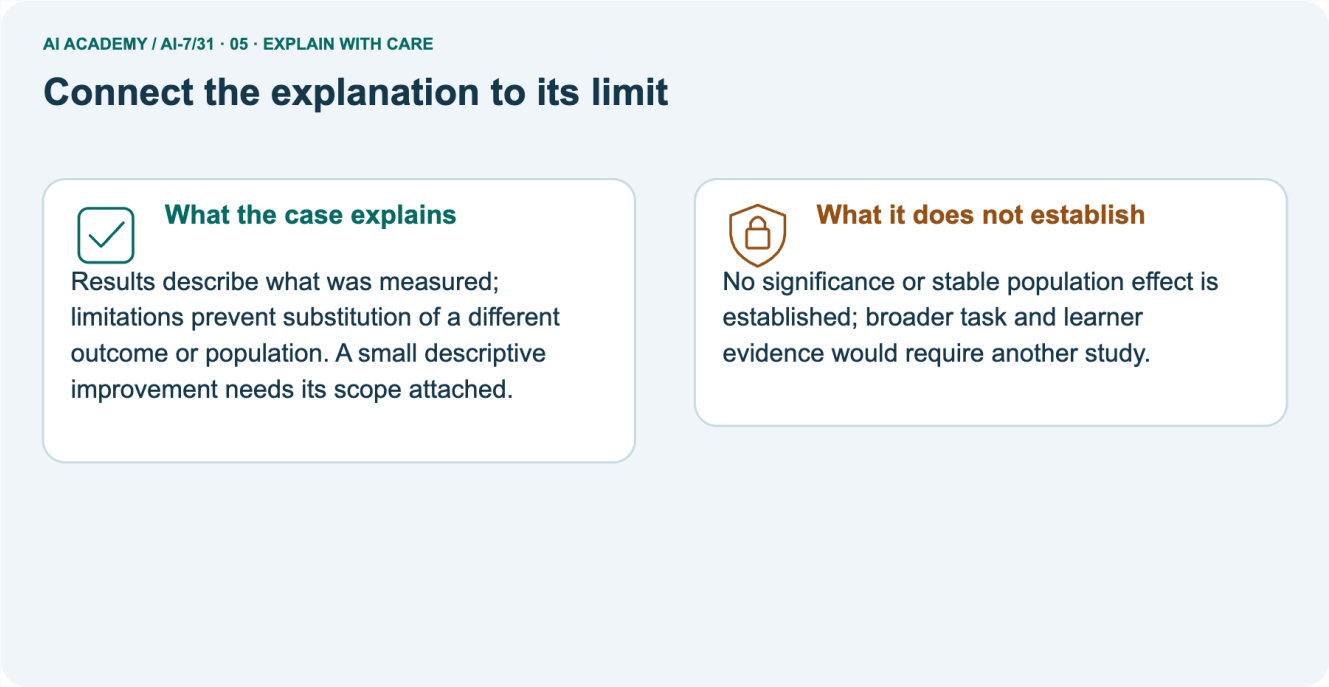
Reason: The measured unit was answers, not student understanding or intelligence.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The measured unit was answers, not student understanding or intelligence.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Results describe what was measured; limitations prevent substitution of a different outcome or population. A small descriptive improvement needs its scope attached.

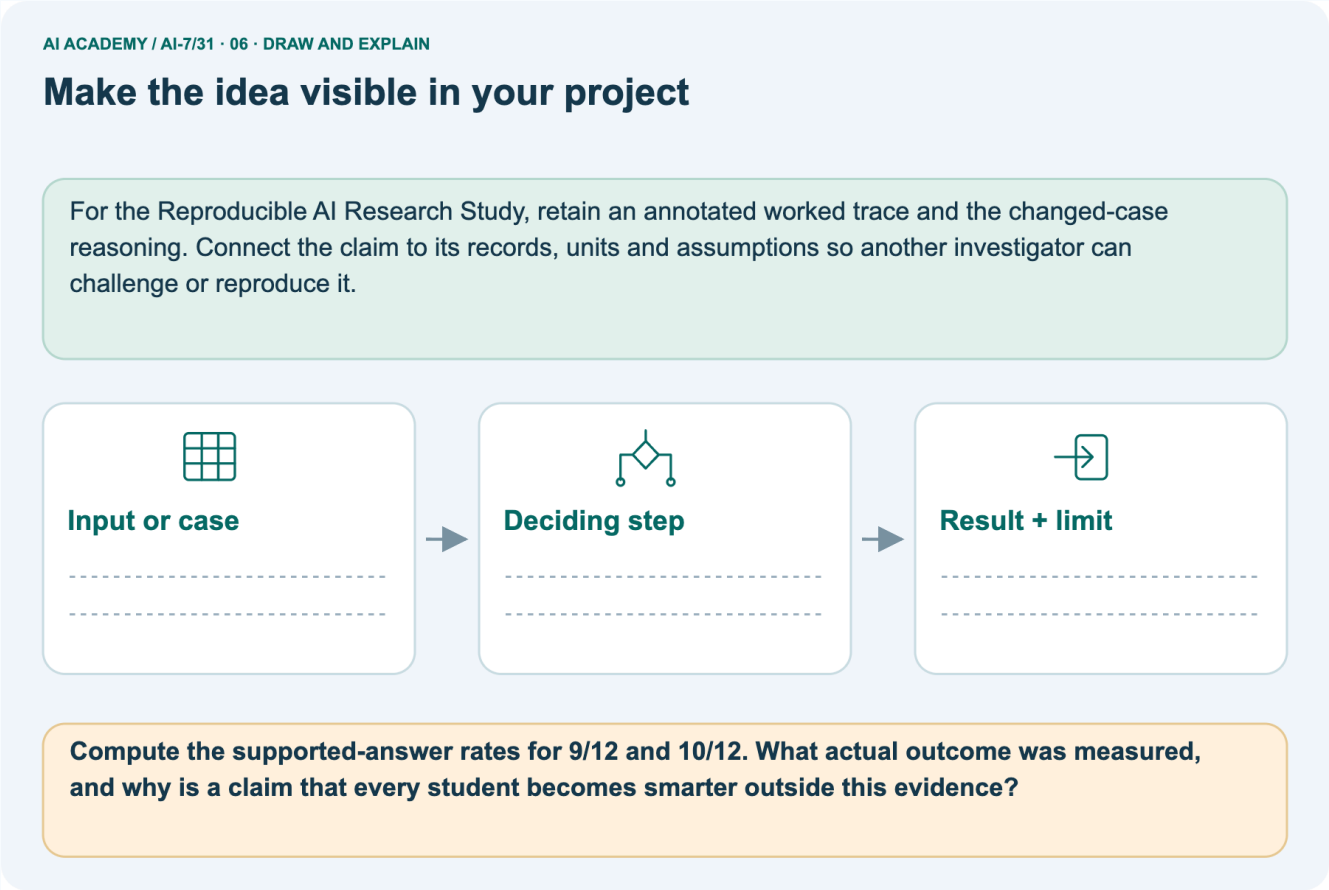
Limit: No significance or stable population effect is established; broader task and learner **evidence** would require another study.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: No significance or stable population effect is established; broader task and learner **evidence** would require another study.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Compute the supported-answer rates for 9/12 and 10/12.

What actual outcome was measured, and why is a claim that every student becomes smarter outside this **evidence**?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: A is 75%; B is about 83.3%, a one-case improvement. Name the collection, rubric and twelve-case denominator. Do not infer that children learn more from this answer-support measure.

Changed case: Remove that sentence as unsupported by the study. The measured unit was answers, not student understanding or intelligence.

### ▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

**01. Writing results and limitations: the case**

**02. Follow the deciding steps**

**03. Supported answers in one collection**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

### ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.